Private Network Connectivity

Microsoft Fabric & Power BI to On-Premises SQL Server

Four reference combinations using ExpressRoute, Site-to-Site VPN, VNet data gateway, on-premises data gateway and managed private endpoints

1. **VNet data gateway + ExpressRoute**
2. **VNet data gateway + S2S VPN**
3. **On-premises data gateway**
4. **Managed private endpoint + PLS**

Architecture options for discussion

Three connectors, two hybrid links, four combinations

First clarify the building blocks: where the connector runs and which Fabric / Power BI workloads it serves

Private path • Gateway placement • Workload coverage

VNet data gateway

WHERE IT RUNS
Runs in YOUR Azure VNet (Microsoft-managed containers in a delegated subnet). It exists only in Azure; there is no on-premises variant.

WORKLOADS
Semantic models, paginated reports, Dataflow Gen2, pipelines, Copy job, Mirroring

ON-PREM LEG
Reaches on-prem SQL through ExpressRoute or Site-to-Site VPN attached to the VNet

On-premises data gateway

WHERE IT RUNS
Installed on Windows servers in YOUR datacenter. This is the 'on-premises gateway' option; it dials out and needs no inbound ports.

WORKLOADS
Power BI semantic models, Dataflow Gen2, pipelines, plus Power Platform

ON-PREM LEG
Outbound-only to Azure Relay; ExpressRoute keeps it off the public internet

Managed private endpoint + Private Link Service

WHERE IT RUNS
Fabric-managed endpoint connects to a Private Link Service in YOUR VNet, fronting an internal load balancer / NAT forwarder.

WORKLOADS
Fabric Data Engineering (Spark notebooks, job definitions) and Eventstream; Fabric capacity (F / Trial)

ON-PREM LEG
ExpressRoute or VPN carries VNet-to-on-prem leg

The 4 combinations in this deck

1 VNet data gateway + ExpressRoute private peering 2 VNet data gateway + Site-to-Site IPsec VPN
3 On-premises data gateway (outbound-only, ExpressRoute optional) 4 Managed private endpoint + Private Link Service over ExpressRoute / VPN

ExpressRoute and VPN are interchangeable hybrid links for patterns 1, 2 and 4; running both gives failover.

Pick the connector by workload first, then the hybrid link by bandwidth, SLA and lead time.

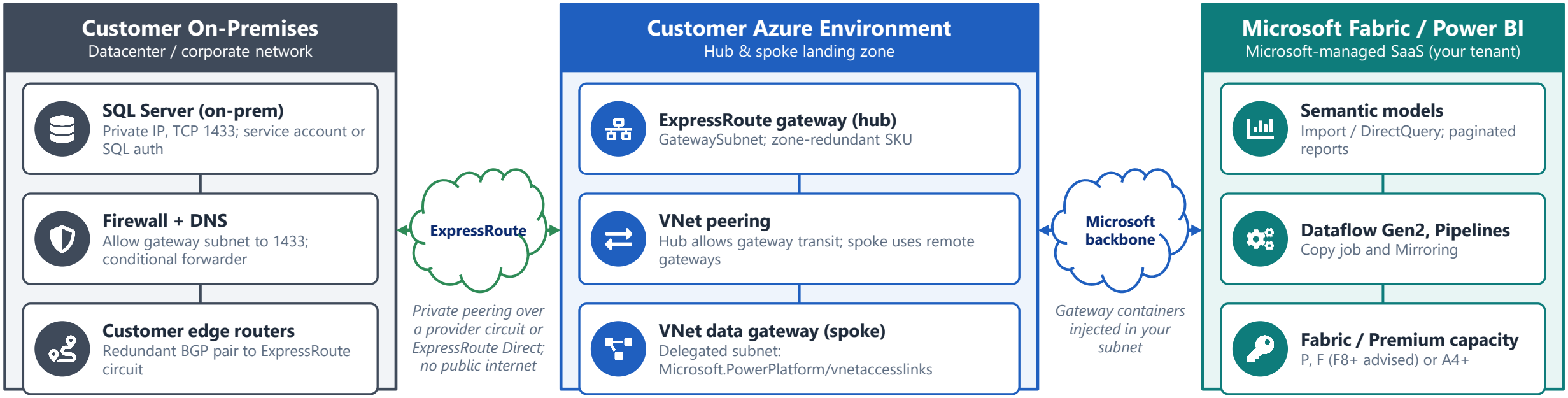
PATTERN 1

VNet Data Gateway over ExpressRoute

Fabric / PBI → On-prem SQL

Private, high-bandwidth path from Fabric / Power BI to on-premises SQL Server

ExpressRoute private peering • Hub & spoke • Managed VNet data gateway



1 Connectivity & Routing

"How does traffic reach on-prem?"

- ExpressRoute circuit lands on the hub ExpressRoute gateway
- Spoke peering uses remote gateways; UDR/NSG must not block on-prem prefixes
- Deterministic latency; traffic stays off the internet

Outcome: Private, predictable path.

2 Gateway & Identity

"Who sets up and uses the gateway?"

- Register Microsoft.PowerPlatform; delegate a dedicated subnet (5 IPs + 1 per member)
- Create gateway in Manage connections and gateways
- Admins grant users access to the data source

Outcome: Governed, shareable gateway.

3 DNS & Data Protection

"How is the data source protected?"

- Resolve the SQL FQDN via Azure DNS Private Resolver or forwarder
- On-prem firewall allows only the delegated subnet CIDR on 1433
- Gateway gives compute isolation and traffic auditing

Outcome: Least-privilege network access.

4 Fit & Watch-outs

"When should we choose this?"

- Best for production: large refreshes and DirectQuery
- Region must be supported; VNet, subscription and RG cannot change after creation
- Circuit lead time and monthly cost

Outcome: Recommended primary pattern.

Pattern 1: Production-grade private path for Power BI and Fabric data integration.

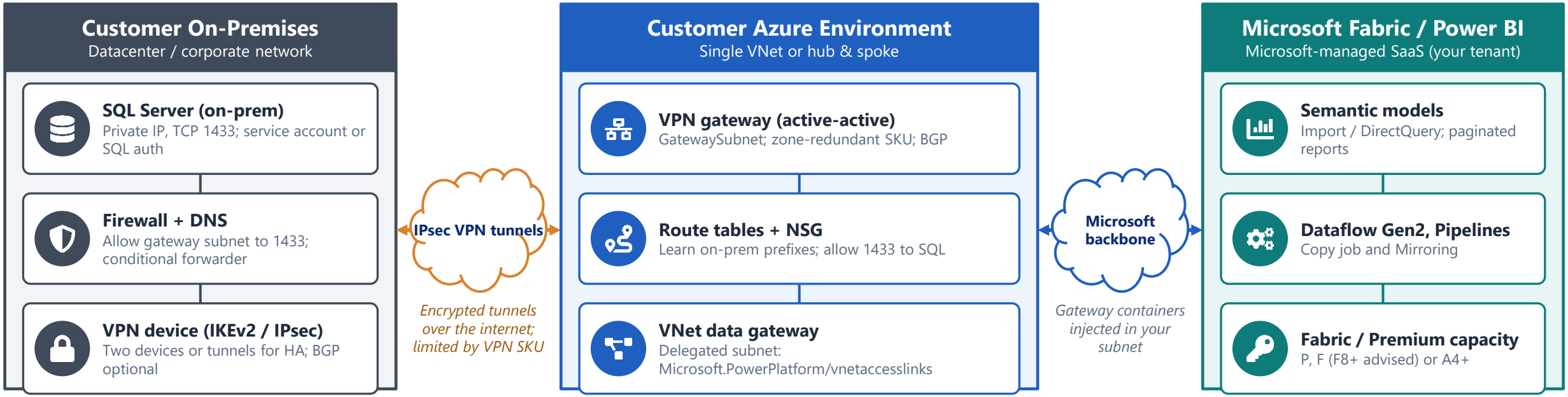
PATTERN 2

VNet Data Gateway over Site-to-Site VPN

Fabric / PBI → On-prem SQL

Encrypted IPsec tunnels as a faster, lower-cost route to on-premises SQL Server

Site-to-Site IPsec / IKE • Active-active VPN gateway • BGP routing



1 Connectivity & Routing

"How does traffic reach on-prem?"

- Two S2S tunnels from an active-active VPN gateway to the on-prem device
- Exchange routes with BGP, or define static prefixes
- Can coexist with ExpressRoute as automatic failover

Outcome: Fast to stand up; resilient.

2 Gateway & Identity

"Who sets up and uses the gateway?"

- Same VNet data gateway setup as Pattern 1
- Delegated subnet must reach the VPN gateway through routing
- Share the data source with Fabric / PBI users

Outcome: Identical gateway experience.

3 DNS & Data Protection

"How is the data source protected?"

- IPsec encrypts the tunnel; keep TLS on the SQL connection too
- Private DNS forwarding for on-prem names
- Monitor tunnel health with Azure Monitor

Outcome: Encrypted, auditable path.

4 Fit & Watch-outs

"When should we choose this?"

- Best for PoC, small refreshes or an ExpressRoute backup
- Internet path means variable latency and throughput
- Size the VPN SKU to peak refresh windows

Outcome: Low-cost start or failover.

Pattern 2: Quickest private route; pair with ExpressRoute for resilience.

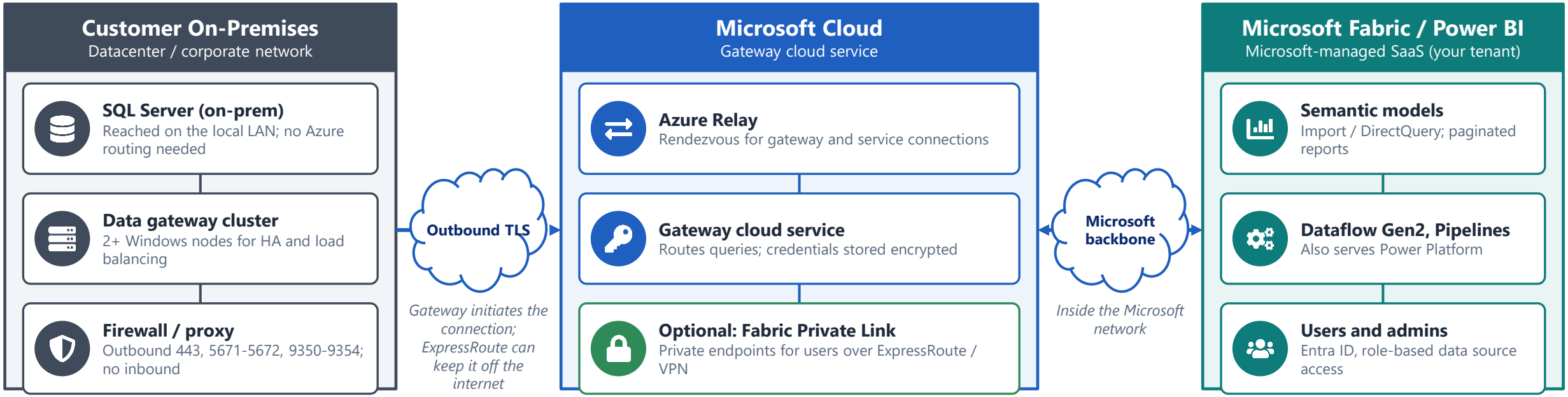
PATTERN 3

On-Premises Data Gateway (Outbound-Only)

Fabric / PBI → On-prem SQL

Gateway runs next to the SQL Server and dials out; no inbound firewall holes

Gateway cluster on-prem • Azure Relay • ExpressRoute optional



1 Connectivity & Routing

"How does traffic reach on-prem?"

- Gateway opens outbound connections to Azure Relay
- ExpressRoute keeps gateway-to-service traffic off the public internet
- Data source is accessed on the LAN

Outcome: No inbound exposure.

2 Gateway & Identity

"Who sets up and uses the gateway?"

- Install on dedicated servers near SQL; register to the tenant
- Cluster nodes for HA; keep the gateway updated
- Gateway admins manage data sources and users

Outcome: Simple setup, you own the hosts.

3 Data Protection

"How is the data source protected?"

- Credentials are encrypted and stored by the gateway service
- Restrict outbound traffic to required endpoints
- Private Link can lock down user access to the service

Outcome: Controlled egress.

4 Fit & Watch-outs

"When should we choose this?"

- Best when no Azure network is wanted, or as a quick start
- You patch, size and monitor the gateway servers
- No VNet-level traffic auditing or Azure private endpoints

Outcome: Lowest network effort.

Pattern 3: The on-premises counterpart to the VNet gateway: simple, outbound-only.

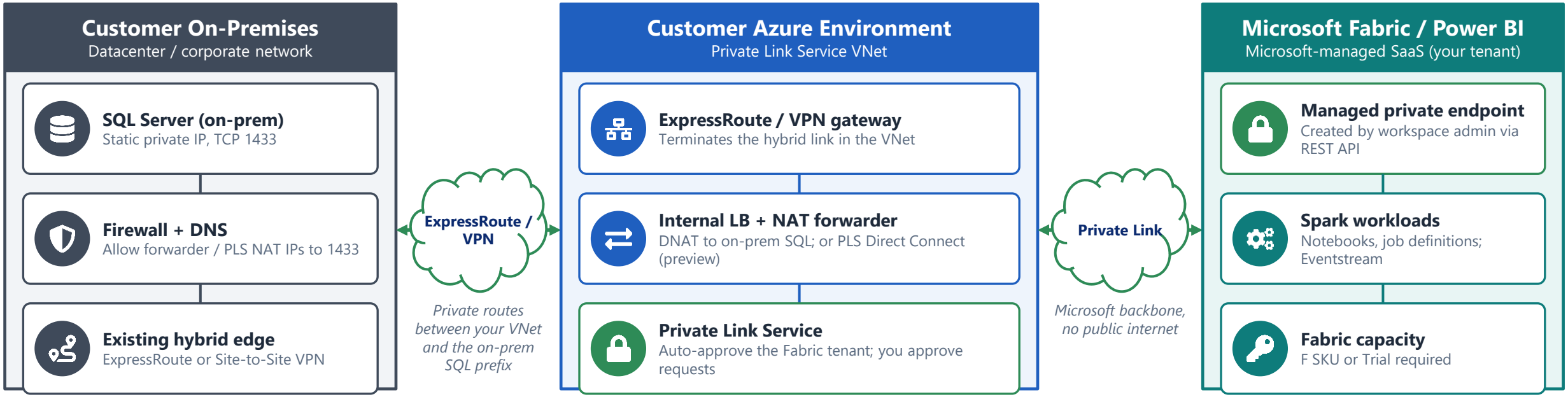
PATTERN 4

Managed Private Endpoint + Private Link Service

Fabric Spark → On-prem SQL

Fabric Spark workloads reach on-premises SQL through your Azure Private Link Service

Managed private endpoint • Private Link Service • ExpressRoute / VPN



1 Connectivity & Routing

"How does traffic reach on-prem?"

- Hybrid link ends in a VNet that hosts the Private Link Service
- Standard ILB with NAT forwarder VMs forwards 1433 to on-prem SQL
- Direct Connect (preview) can target a static IP without the LB

Outcome: Private, gateway-free path.

2 Fabric & Identity

"Who creates and approves it?"

- Workspace admin creates the endpoint with the Managed Private Endpoints REST API
- Network admin approves the pending connection on the PLS
- Connect from Spark with the FQDN

Outcome: Two-party approval.

3 Data Protection

"How is the data source protected?"

- Traffic stays on the Microsoft backbone
- NSGs limit 1433 to the forwarders; admin ports only from trusted ranges
- Use customer-managed keys for data at rest

Outcome: Isolated, approved access.

4 Fit & Watch-outs

"When should we choose this?"

- Best for Spark and lakehouse ingestion without a gateway
- You run the NAT VMs, or accept a preview feature
- Not for semantic models or paginated reports

Outcome: Complements Patterns 1 or 2.

Pattern 4: Gateway-free private path for Fabric Data Engineering workloads.

Side-by-side comparison

Pick by workload, operational ownership and bandwidth needs

	1 VNet GW + ExpressRoute	2 VNet GW + S2S VPN	3 On-prem data gateway	4 Managed PE + PLS
Hybrid link	ExpressRoute private peering	IPsec VPN over internet	Outbound 443 / AMQP to Azure Relay; ExpressRoute optional	ExpressRoute or VPN to PLS VNet
Connector runs in	Your Azure VNet (Microsoft-managed)	Your Azure VNet (Microsoft-managed)	Your on-prem servers	Fabric managed VNet to your Private Link Service
Workloads	Semantic models, paginated, Dataflow Gen2, pipelines, Copy job, Mirroring	Same as pattern 1	Semantic models, Dataflow Gen2, pipelines, Power Platform	Spark notebooks and job definitions, Eventstream
Public internet	None	Encrypted tunnel crosses internet	Outbound only; none with ExpressRoute	None
You operate	ExpressRoute circuit, hub and spoke, gateway subnet	VPN devices, VPN gateway, subnet	Gateway servers, patching, cluster	ILB and NAT VMs (or preview), approvals
Main watch-out	Lead time and cost; supported regions	Variable latency and throughput	Hardware sizing; no VNet-level audit	REST-only steps; not for semantic models
Best fit	Production, high volume	PoC or backup path	Quick start, no Azure network	Spark ingestion, gateway-free

Patterns can be combined: for example 1 for Power BI and pipelines, plus 4 for Spark.

Recommended approach and next steps

Primary
Pattern 1: VNet data gateway over ExpressRoute for Power BI semantic models and Fabric data integration.

Resilience
Pattern 2: add a Site-to-Site VPN on the same VNet as an ExpressRoute failover, or as the PoC path.

Spark ingestion
Pattern 4: managed private endpoint + Private Link Service on the same hybrid link.

Fallback
Pattern 3: on-premises data gateway when Azure networking is not yet ready.

Next steps

1. Confirm workloads: semantic models, Dataflow Gen2, pipelines, Spark
2. Confirm Fabric region and an Azure region on the VNet gateway supported list
3. Choose ExpressRoute circuit, VPN, or both, and size bandwidth
4. Design hub and spoke, delegated subnet, DNS and firewall rules
5. Pilot with one on-prem SQL source and validate refresh, DNS and failover

Sources: Microsoft Learn - What is a virtual network data gateway; Create virtual network data gateways; Use virtual network data gateway and data sources in Power BI; Connect to on-premises data sources using managed private endpoints; Overview of managed private endpoints for Fabric; On-premises data gateway architecture; ExpressRoute circuits and peering.